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Inference Problem Set

1) $\bar{X} = 52$, $S = 10$, $n = 100$

- Standard error $SE(\bar{X}) = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$

$H_0 : \mu = 50$, $H_a : \mu \neq 50$

$$z = \frac{\bar{X} - \mu}{SE(\bar{X})} = \frac{52 - 50}{1} = \frac{2}{1} = 2$$

at:

$\alpha = 0.10$:	$\lvert z \rvert \geq 1.645$?	yes
$\alpha = 0.05$	$\lvert z \rvert \geq 1.96$?	yes
$\alpha = 0.01$	$\lvert z \rvert \geq 2.576$?	no

We <u>fail to reject</u> the null hypothesis that that the population mean is 50 at a <u>0.01 significance level</u>, <u>but</u> <u>we reject</u> the null hypothesis at a significance level of <u>0.05 and 0.10</u> which gives us significant evidence that the population mean is not 50.

95% CI

$$\bar{X} \pm 1.96 * SE(\bar{X})$$

$52 + (1.96 * 1) = 53.96$

$52 - (1.96 * 1) = 50.04$

(50.04, 53.96) ← 95% confidence interval

2) $\hat{S} = 3.1$, $\widehat{SE}(\hat{S}) = 1.3$

90%: $3.1 + (1.645 * 1.3) = 3.1 + 2.1385 = 5.2385$
$3.1 - (1.645 * 1.3) = 3.1 - 2.1385 = 0.9615$

95%: $3.1 + (1.96 * 1.3) = 3.1 + 2.548 = 5.648$
$3.1 - (1.96 * 1.3) = 3.1 - 2.548 = 0.552$

99%: $3.1 + (2.576 * 1.3) = 3.1 + 3.3488 = 6.4488$
$3.1 - (2.576 * 1.3) = 3.1 - 3.3488 = -0.2488$

$H_0: S = \emptyset$ vs $H_a: S \neq \emptyset$

90% CI: $(0.9615, 5.2385)$, $\alpha = 0.1$
95% CI: $(0.552, 5.648)$, $\alpha = 0.05$
99% CI: $(-0.2488, 6.4488)$, $\alpha = 0.01$

We reject the null hypothesis at a significance level of 0.1 and 0.05, but we fail to reject at a significance level of 0.01 because the interval contains $\emptyset$

3) $\bar{x} = 11.5$, $s = 3$, $n = 36$

$Z = \frac{\bar{x} - \mu}{SE(\bar{x})} \rightarrow SE(\bar{x}) = s/\sqrt{n} = 3/\sqrt{36} = 3/6 = 1/2$

$\frac{11.5 - 12}{0.5} = \frac{-0.5}{0.5} = (-1)$ $\quad |-1| \geq 1.96$? no

- 95% CI: $11.5 \pm (1.96 * 1/2) \rightarrow (10.52, 12.48)$
- p value at $Z = 1/2 \rightarrow \approx (1 - 0.7088) * 2 = 0.58$
- test at $\alpha = 0.05$: $0.22 \leq 0.05$? no
- Z score test: $|-1| \geq 1.96$? no

conclusion: we fail to reject the null hypothesis that the population mean is 12, meaning our test implies 12 is a plausible value for the population mean.

4) $\bar{X}_A = 25 \quad S_A = 8 \quad n_A = 100$

$\bar{X}_B = 22 \quad S_B = 7 \quad n_B = 100$

$$Z = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}}} = \frac{25-22}{\sqrt{\frac{8^2}{100} + \frac{7^2}{100}}} = \frac{3}{\sqrt{.64+.49}} = \frac{3}{\sqrt{1.13}}$$

≈ 2.822

95% CI: $3 \pm \left(1.96 * \sqrt{\frac{8^2}{100} + \frac{7^2}{100}}\right) =$

$= 3 \pm (1.96 * \sqrt{1.13})$

$= 3 \pm 2.0835$

$\Rightarrow (0.9165, 5.0835)$

5) $\bar{X}_A = 14, \quad S_A = 6, \quad n_A = 81$

$\bar{X}_B = 12, \quad S_B = 5, \quad n_B = 64$

Standard error: $\sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}}$

≈ 0.914

Test statistic

$$Z = \frac{(X-Y) - 0}{SE(X-Y)} = \frac{2}{0.914} \approx 2.188$$

6) $\bar{X}_A = 3.2, \quad S_A = 1.5, \quad n_A = 49$

$\bar{X}_B = 2.8, \quad S_B = 1.4, \quad n_B = 64$

Test statistic: $Z = \frac{3.2 - 2.8}{\sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}}} \approx \frac{0.4}{0.2767} \approx 1.446$

Two sided p value: $(1 - 0.9265) * 2 = 0.141$

We fail to reject the null hypothesis that there is no difference in defect stores across the two factories at a significance level of 0.05. The difference is NOT statistically large relative to the noise.